Wk 3 Supervision Problems 2

Luca Bollini

23 October 2026

Difference Equations

1. Find the nth term of the Fibonacci sequence.

2. Let us assume that
$$\frac{2x+7}{x^2+5x+6}$$
can be written as an expansion in ascending powers of x in the form $a_0 + a_1x + a_2x^2 + \dots$. Prove that for $n \geq 2$
$$a_n + 5a_{n+1} + 6a_{n+2} = 0$$
and solve the difference equation.

3. A plant is such that one year after being planted as a seed, it produces 21 new seeds, and each year thereafter produces 44 new seeds. Assume that as soon as a new seed is produced it is immediately planted. Taking u_n to be the number of seeds produced in the nth year, explain why
$$u_{n+1} = 21u_n + 44(u_0 + u_1 + \cdots + u_{n-1}).$$
Hence show that
$$u_{n+2} - 22u_{n+1} - 23u_n = 0$$
and solve the difference equation.

Null Sequences

1. Prove that $|a_n| \to \infty$ if and only if $\frac{1}{a_n} \to 0$.

2. Prove that the sequence
$$b_n = \frac{n}{n^2+1}$$
is a null sequence.